卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-SG-0204 30-B003701S-S0204

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CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

水布 专业 第 02 卷 第 04 册
Hydraulic SPECIALITY VOLUME 02 BOOK 04

卷册名称 深井泵房及补充水管道布置图
Volume name Layout Drawing Of Deep Well And Make-up Water Piping

批 准 APPROVED BY	2020.2.25
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	2020.2.25
图 纸 5 张 DRAWING NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-S0204-001	深井泵房安装说明	B	1		
		Installation Instructions For Deep Well Pump House				
2	30-B003701S-S0204-002	深井与补充水管平面布置图	B	1		
		Deep Well And Make-up Water Pipe Layout Drawing				
3	30-B003701S-S0204-003	#1深井泵房布置图	B	1		
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4	30-B003701S-S0204-004	#2深井泵房布置图	B	1		
		#2 Deep Well Pump House Arrangement Drawing				
5	30-B003701S-S0204-005	深井泵房及补充水管材料表	B	1		
		Bill Of Material For Deep-well pump House And Make-up Water Pump				

注:

1. 本卷册为深井及深井泵房内的设备、工艺管道安装图，站区补充水管道布置图。本工程共设2口水井及深井泵房，均由土建施工单位完成。
根据《供水水文地质勘察报告》，拟建站址区除表层2~3m为粉质黏土层外，其余地层均为粉细砂，粉细砂底板埋深超过80m，地下水位埋深约5m，
含水层为潜水，厚度较大，水量丰富。本工程试验抽水井深为80m，直径为406.4mm，桩其中地面上0.5m到地下12.0m安装φ254mm侧壁不开孔的
玻璃纤维套管，12.0~74.0m安装φ254mm玻璃纤维滤水花管，74.0~80.0m安装沉淀管，沉淀管底用止水材料封闭，井管采用钻机垂直提吊下入孔内，
在沉淀管底部加装一组扶正器，使井管位于钻孔中心。抽水井7.0m以下将选配好的砾砂采用动水方式均匀投入井管环状间隙，0.5~7.0m采用用直径2~3cm
优质黏土球进行止水密实回填，最后改用石子固井至地面。根据上述勘察结果，本工程在厂区范围内设置两口抽水井，抽水井间距约300m，抽水井静水位
最大埋深6m，开采井群抽时水位降5m，由于目前深井泵尚未招标，暂按水泵进水口淹没深度在动水位以下20m计，水泵进水口设置在埋深约31m。

2. 深井施工
2.1 由于成井施工的特殊性，为保证施工质量，要求施工队伍为具有丰富经验的专业队伍，最好有当地施工经验。施工前请详细了解
《供水水文地质勘察报告》，根据含水层分布及特性，尤其注意粉细砂层的影响，选择合适的过滤器。
2.2 永久性深井，应充分洗井并做抽水实验，要求水井设计出水量不小于80 m³/h，并对此试验结果进行推算，在水量最枯时也应满足此水量要求。洗井后应保证出水
含砂量应小于1/200000（体积比）。图中2口井的位置坐标为暂定，如实际工程井的出水流量和含沙量不满足设计要求，应另行选址打井。
2.3 出水量及含砂量检测方法及其它要求遵照国家标准《管井技术规范》（GB50296-2014）的规定。
2.4 水井施工完毕，应提供成井资料（如抽水实验成果报告、水井综合柱状图、成井深度、静水位、动水位、含水层分布等）。
2.5 水井施工完毕，还应提供水质检验报告。水质检验报告应由取得业主和监理同意的卫生防疫部门出具，如果在国内检测，应由县
级及以上卫生防疫部门出具，水样的提取和送检均应在卫生防疫部门或其指导下进行，以免其它因素影响水质。水质检验项目
应包括感官性状和一般化学指标、微生物指标、毒理指标。

3. 深井潜水泵运行与安装
3.1 深井潜水泵启动前,应检查自动排气阀是否灵活好用。
3.2 深井潜水泵运行与生产消防水池及生活水箱水位联锁，水池高水位（A.L）停泵，低水位（L.L）启泵。
3.3 深井潜水泵一运一备，且定时自动切换运行。
3.4 深井潜水泵的运行、故障状态信号上传给站控系统。
3.5 上述自动联锁控制及信号远传功能由深井泵控制柜实现，同时应含下列控制和信号显示功能：总开关（自动、手动、停机切换）、水泵手动控制（启动、停止）、
应有深井泵运行、停止、故障、检修等信号显示、电机防干烧功能，控制柜置于深井泵房内。
3.6 深井潜水泵控制柜以及从控制柜至深井泵的电缆由深井潜水泵厂家配套供货。
3.7 当深井潜水泵安装检修或停止运行较长时间后,再启动时须先排除井内污染水,
待深井水质无变化再送入出水管网。

4. 管道
4.1 材质: 泵后的给水管道采用衬塑镀锌钢管。
4.2 连接: 衬塑镀锌钢管采用丝扣连接，管道与阀门之间采用法兰连接。

5. 施工与验收
5.1 施工与验收参照执行以下国家现行规范:
1）《管井技术规范》（GB50296-2014）
2）《给水衬塑符合钢管》CJ/T 136-2007
3）《给水衬塑可锻铸铁管件》CJ/T 137-2008
4）《钢结构工程施工质量验收规范》GB50205-2001
5）《电力建设施工及验收技术规范 火力发电厂焊接篇》DL869-2012
6）《给水排水管道工程施工及验收规范》GB50268-2008
7）《电力建设施工及验收技术规范-管道篇》DL5190.5-2012

1. This volume shows equipments & process piping installation in deep well pump house and make up water piping layout.Two wells and
deep well pump houses shall be completed by civil engineering construction company.
According to <Hydrogeological Survey Report>,except surface layer of 2~3m silty clay,other layers are made up of silty sand in station site.
Buried depth of silty sand bottom is over 80 m,The embedded depth of groundwater is about 5 m,aquifer is phreatic water with large thickness and rich water quantity.
The test Wells have a depth of 80 m and diameter of 406.4 mm.φ254mm glass fibre sleeve without holes in the side wall is installed from 0.5m above the ground to 12m
underground.φ254mm glass fiber filter water pipe is installed between 12m~74m underground.Sedimentation pipe is installed between 74m~80m underground.
Bottom of sedimentation pipe is closed by stop material.The well tube is hanged down vertically into hole by drill,adding a set of centralizer at the bottom of the tube,to
make tube located in drilling center.Below 7m of pumping well will be equipped with good gravel sand that is sent into well tube annular gap by way of dynamic water,
Dense backfill shall be conducted with high quality clay ball with a diameter of 2~3cm,finally using stones to cement well from 0.5m underground to the ground.
According to results of the survey, this project shall set two pumping Wells within the scope of the station,pumping well has a spacing of about 300m,
Maximum depth of static water level in pumping well is 6m,water level drop is 5m at the time of pumping.Bid inviting of deep-well pump is not carried out,
submerged depth of pump inlet is ~20m under dynamic water level temporarily, water pump inlet is installed at about 31m below underground.

2. Construction of deep well
2.1 As particularity of well competation,in order to ensure construction quality,The construction team must be a professional team with rich experience, preferably with local
construction experience.Please study <Hydrogeological Survey Report> in detail before construction,based on distribution & characteristics of the aquifer,pay particular
attention to influence of the fine sand layer,select appropriate filters.
2.2 Permanent deep well should be fully washed and carried out pumping experiment,design water yield shall not be less than 80 m³/h,and the test results shall be calculated
to ensure that water yield meet the requirements at the driest season and sand content is less than 1/200000(volume ratio).Location coordinates of 2 wells in the figure is
tentative,if actual water yield and sand content is not qualified,well drilling should select another new site.
2.3 Detection method and other requirements of hydraulic discharge & sand content must comply with national standard《Technical code for tube well》(GB50296-2014).
2.4 After well construction is completed,following well completion data shall be provided:(Such as the pumping experiment results report,
well integrated bar chart, well depth, static water level, dynamic water level, aquifer distribution, etc.)
2.5 After well construction is completed,water quality monitoring report must be provided.Water quality monitoring report should be issued by the health and epidemic prevention department
that CET and supervision company are acceptable.If monitoring work conducted in china,health and epidemic prevention department should be at and above the country level,
The extraction and inspection of water samples should be carried out under the guidance of the department of health and epidemic prevention, so as to prevent other factors from
affecting the water quality.The water quality inspection items should include sensory properties & general chemical indicators, microbial indicators and toxicological indicators.
3. Operation and installation of deep well submersible pump
3.1 Before starting pump,automatic air valve should be checked to make sure that it is flexible and easy to use.
3.2 Operation of deep well pump shall be interlocked with water level of industrial water pond,fire water pond and
domestic water tank,pump stops at high water level(A.L),starts at low water level(LL).
3.3 One pump is in use and the other one is standby,two pumps shall be in automatic switching operation.
3.4 Running and fault status signal of deep well should be uploaded station control system.
3.5 Automatic interlocking control and signal remote transfer mentioned above shall be all realized by control cabinet,following control and signal display function should be
included：main switch(automatic,manual,shutdown switch),manually control of pump(start,stop);operation,stop,fault,maintanance signal of pump
shall be displayed,and motor dry-boiling prevent function shall be equipped,control cabinet shall be set in deep well pump house.
3.6 Control cabinet and cable from the control cabinet to the deep well pump shall be supplied by pump manufacturer.
3.7 If deep-well pump has been through a long time of maitanance or shutdown,polluted water in the well must be drawn off as pump restarting,
only when water quality of deep well is good and unchanged,it could be sent to oulet pipe.
4. Piping
4.1 Material: outlet pipe of pump adopts galvanized steel pipe of lining plastic.
4.2 Connection: galvanized steel pipe of lining plastic adopts screwed connection,
and pipe and valve adopts flanged joint.
5. Construction and acceptance
5.1 Construction and acceptance shall comply with currently national codes as followed:
1）《Technical code for tube well》(GB50296-2014)
2）《Steel pipes of lining plastic for water supply》CJ/T 136-2007
3）《Malleable iron threaded fitting of lining plastic for water supply》CJ/T 137-2008
4）《Code for acceptance of construction quality of steel strucrures》GB50205-2001
5）《The code of welding for power plant》DL869-2012
6）《Code for construction and acceptance of water and sewerage pipeline works》GB50268-2008
7）《Technical specification for thermal power erection and construction part 5:piping and system》DL5190.5-2012

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 4 - 0 0 1 B

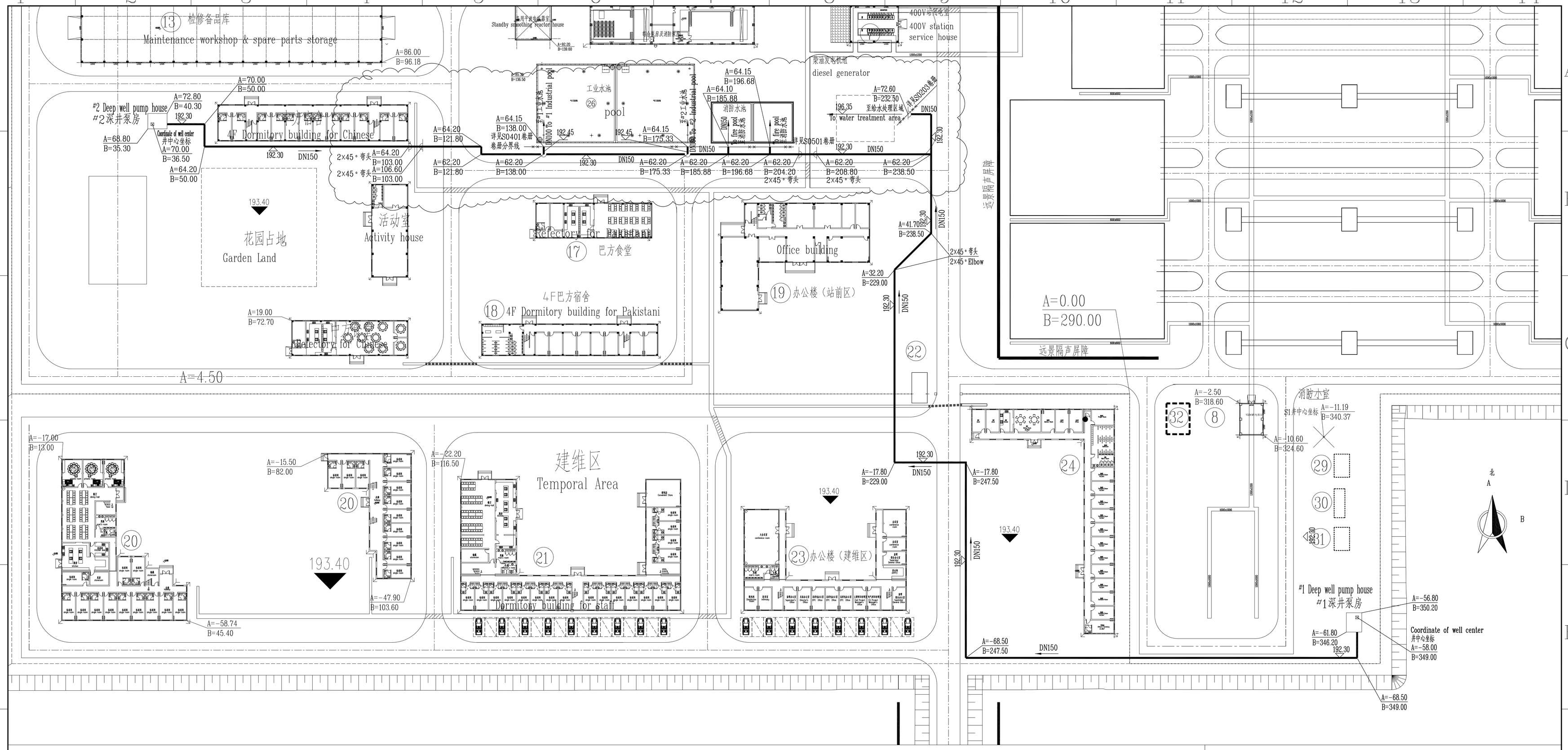
	姓名 Name	日期 Date
设计 Designed	谭叶青	2020.2.25
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

深井泵房安装说明
Installation Instructions For Deep Well Pump House

30-B003701S-S0204-001

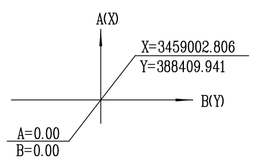
版本 REV. B

A1(594x841)



说明:

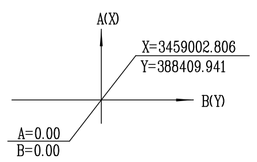
1.本图高程采用EGM2008高程系统，坐标采用站区建筑（A、B）坐标系统，与巴基斯坦UTM-43N平面坐标系关系如下所示:



2.图中高程以及平面坐标以米（m）为单位，尺寸以毫米（mm）为单位。
3.场地平均设计标高为193.40m，站内道路为郊区型道路，道路中心标高193.60m。
4.由于地下水处理工艺未确定，补充水管的布置方案为暂定。
5.结合水文地质勘测资料及总平面布置要求，确定了深井的位置，以满足现场施工进度需要，深井的构造及施工要求详见《水文地质勘测报告》。
6.其他说明详见本卷册01图。

Notes:

1.EGM2008 elevation is adopted in this volume,The station coordinate(AB) is adopted in this volume,station coordinate and UTM-43N system conversion formula shows as follow:



2.Unit of coordinate and elevation is m,that of dimensions are in mm.
3.Average elevation of outdoor ground is 193.40m.Station road adopts suburban road type,elevation of road center is 193.60m.
4.As the groundwater treatment process is uncertain, the arrangement of make-up water pipe is tentative.
5.Combined with hydrological geological survey data and the general layout requirements,position of deep well is determined, in order to come up with construction progress,construction requirements of deep well refer to the hydrogeological survey report.
6.Other descriptions refer to S0204-01 drawing of this volume.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 4 - 0 0 2 B

	姓名 Name	日期 Date
设计 Designed		2020.2.25
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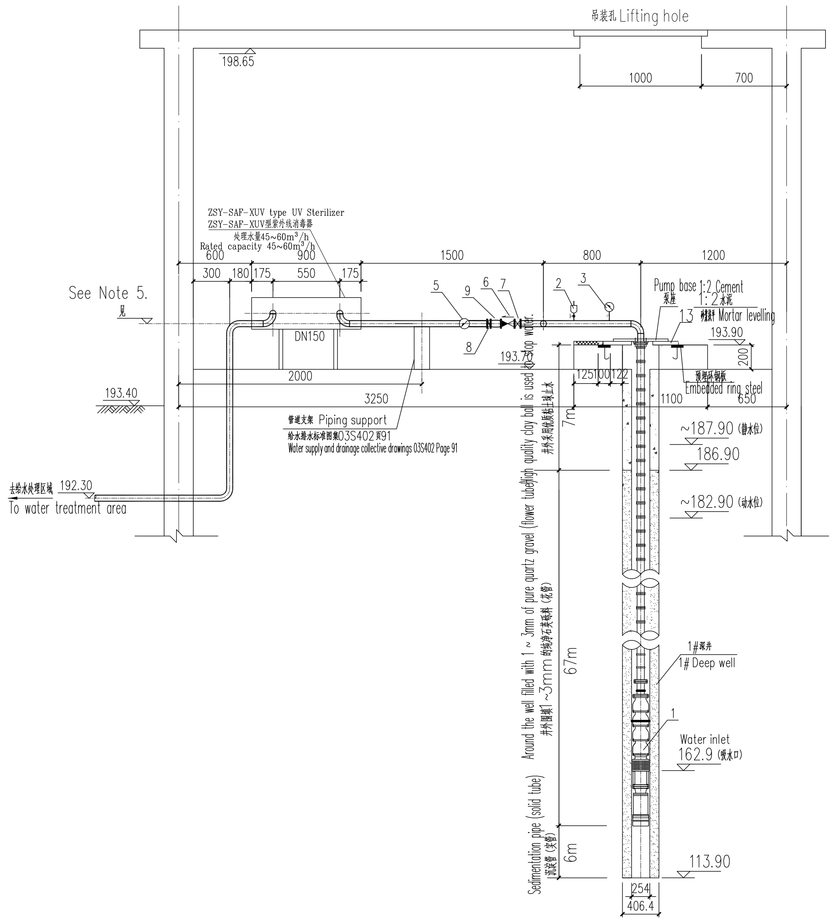
深井与补充水管平面布置图
Deep Well And Make-up Water Pipe Layout Drawing

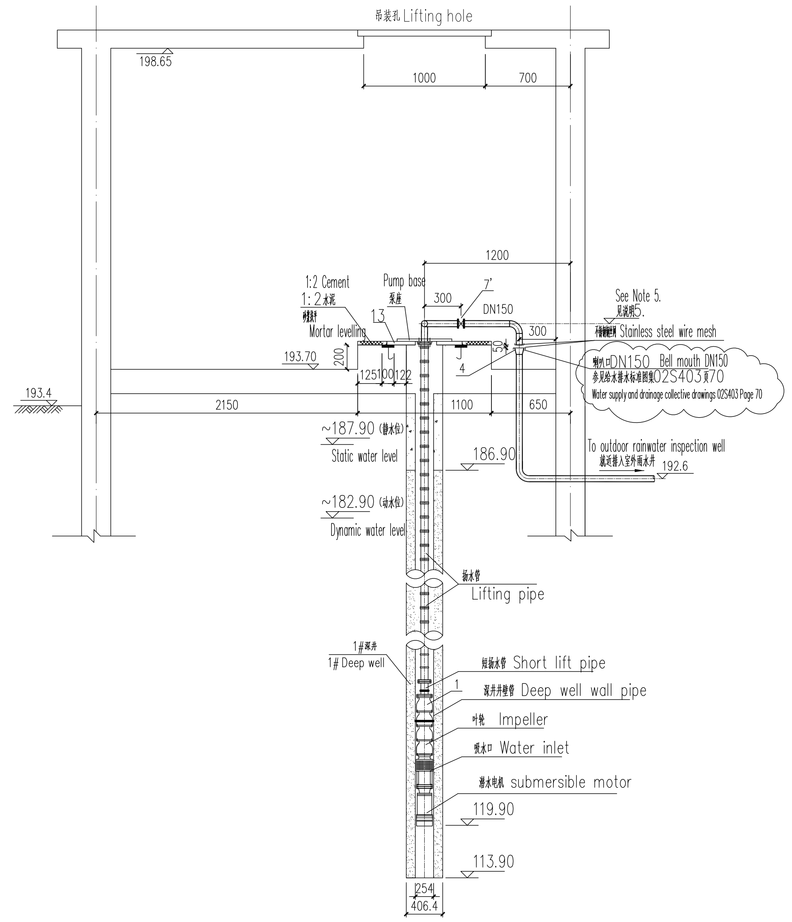
比例 Scale: 1:500 | 图号 DWG NO.: 30-B003701S-S0204-002 | 版本 REV.: B

594X841



2-2 剖面图

2-2 Cross section view

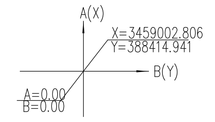


1-1 剖面图

1-1 Cross section view

说 明:

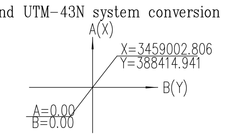
1. 图中坐标和标高单位以m计,其余均以mm计。
2. 本图采用EGM2008高程系统,图中标高为绝对标高。
   本图采用建筑(A、B)坐标,与巴基斯坦UTM-43N平面坐标系关系如下所示:



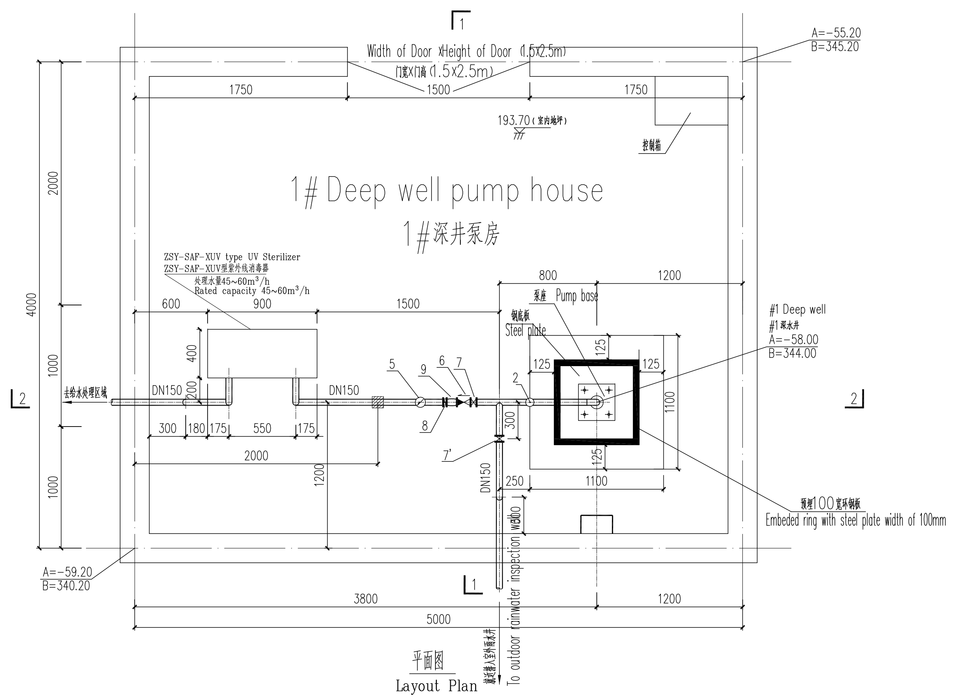
3. 深井施工单位应提供成品井,井壁管管径按ø254×10优质无缝钢管考虑,最终管径、壁厚以勘测专业为准。在实际成井后,应提供资料给设计单位核实深井潜水泵组参数。
4. 混凝土基础及环形埋铁在打井完成后先做好,钢底板待深井潜水泵到货后根据井下安装的最小空间要求确定中间开孔尺寸ø,根据泵座的要求确定螺栓位置(a)和规格(见加工简图)。钢底板加工完毕后现场与环形埋铁焊接,然后将基础四周用1:2水泥砂浆找至同钢底板顶面平。
5. 深井泵出水管中心标高,根据深井泵配带的出水口的安装高度现场确定。
6. 平时正常运行时阀门启闭状态为:阀7常开,阀7'常闭。深井潜水泵安装检修或停止运行较长时间后,再启动时须先放掉井内污染水。操作顺序为:关闭阀7,打开阀7',启动深井潜水泵,运行一段时间后见该管流出的水水质清澈后立即将各阀门恢复到正常运行状态。
7. 深井泵房内设备均需接地。
8. 由于本工程初步设计尚未审定,未通过NTDC审核,最终方案还存在修改的可能性,施工前请联系业主与设计确认。由于目前深井泵设备尚未招标,本图为根据给水排水手册设备资料暂估,深井泵设备应以最终招标资料为准,并在设备到货安装前复核基础及埋件尺寸。
9. 其他说明详见本卷册01图。

Notes:

1. Unit of coordinate and elevation is m,that of other dimensions are in mm.
2. EGM2008 elevation is adopted in this volume,elevation represents absolute elevation.The station coordinate(AB) is adopted in this volume,station coordinate and UTM-43N system conversion formula shows as follow:

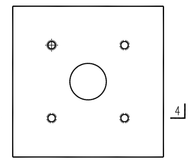


3. Construction contractor shall provide deep well,casing pipe uses ø254×10 high grade seamless steel pipe,final diameter,wall thickness.shall be based on surveying major.After deep well is completed,information of deep-well pumps shall be provided to design institute to check parameters.
4. Concrete foundation and circular embeded part shall be done first after drilling work finished,hole size of steel plate will be determined based on minimum installation space of deep-well pump,bolt location(a) and specification(b) would be determined based on pump frame requirements(fabrication skematic diagram)Steel plate fabrication,after fabricated,will be welded with cicular embeded part on site,then foundation surrounds will be made up level to to top elevation of steel plate with 1:2 cement mortar.
5. Center elevtion of outlet pipe shall be adjusted with actual elevation of autual arrived pump outlet.
6. Opening and closing position in operation : valve 7 normally open , valve 7' normally closed.If deep-well pump going through a long time of maitanance or shutdown polluted water in the well must be drawn off as pump restarting.operation order: close valve 7,then open valve 7' , start deep-well pump,recover the valves to a normal state immediately,provided water quality is clear after running for a while.
7. All equipments in deep-well pump house must be grounded.
8. As basic design has not been reviewed yet by NTDC,final scheme is up on air.Please contact CET and Design Institute before construction in order to make sure that scheme is right.Bid inviting of deep-well pump is not carried out,pump data refer to equipment information of water supply and drainage handbook,deep-well pump equipment shall be subject to final biding informationand review foundation and embeded size before equipment transported to the site.
9. Other descriptions refer to S0204-01 drawing of this volume.

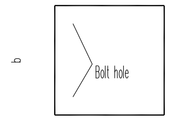


平面图

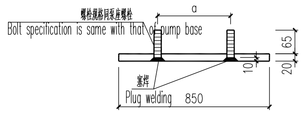
Layout Plan



Processing schematic of Steel bottom plate (thickness 20)

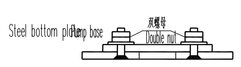


泵座示意图



钢底板加工4—4剖面图

4-4 Cross section drawing for processing of steel bottom plate



Connection layout of pump base and steel bottom plate

13	Steel plate 钢底板	850x850x20
12	90° Elbow 90°弯头	DN150
11	Tee joint 三通	DN150
10	Steel Pipe lined with plastics 衬塑镀锌钢管	DN150 , PN16
9	Flexible rubber joint 可曲挠橡胶接头	DN150 , PN10
8	Butterfly valve 蝶 阀	DN150 , PN10
7(7')	Sluice valve 闸 阀	DN150 , PN10
6	Microresistance slow closing check valve 微阻缓闭止回阀	DN150 , PN10
5	Electromagnetic flowmeter 电磁流量计	DN150
4	Horn mouth 喇叭口	DN150
3	Pressure gage 压力表	Y-60,0~1.0MPa
2	Automatic air valve 自动排气阀	ARSX-0015 DN15
1	Deep-well pump 深井潜水泵组	Q=80m³/h,H=52m,380V , 18.5kW
编号 No.	名 称 Name	规 格 Parameters

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦联带至里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

NLHVDC-LCS-ECI-DD-SG-0204-003 B

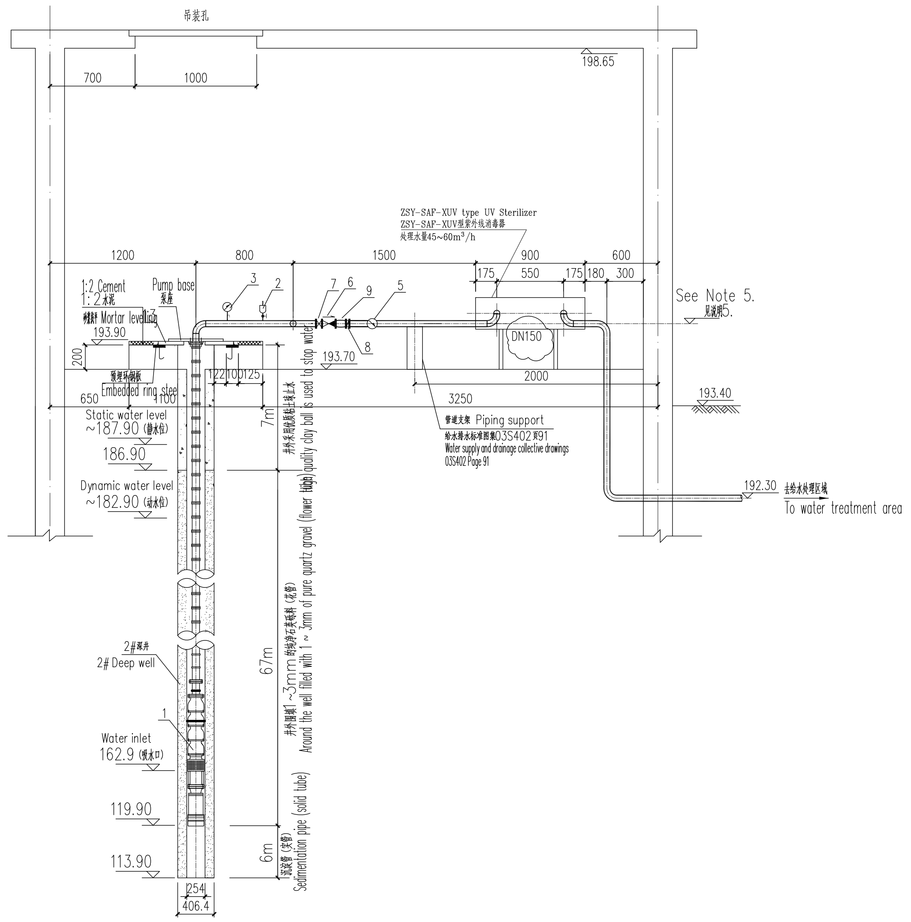
	姓名 Name	日期 Date
设计 Designed	罗双琦	2020.2.25
校核 Checked		
审核 Reviewed		
批准 Approved		

#1深井泵房布置图

#1 Deep Well Pump House Arrangement Drawing

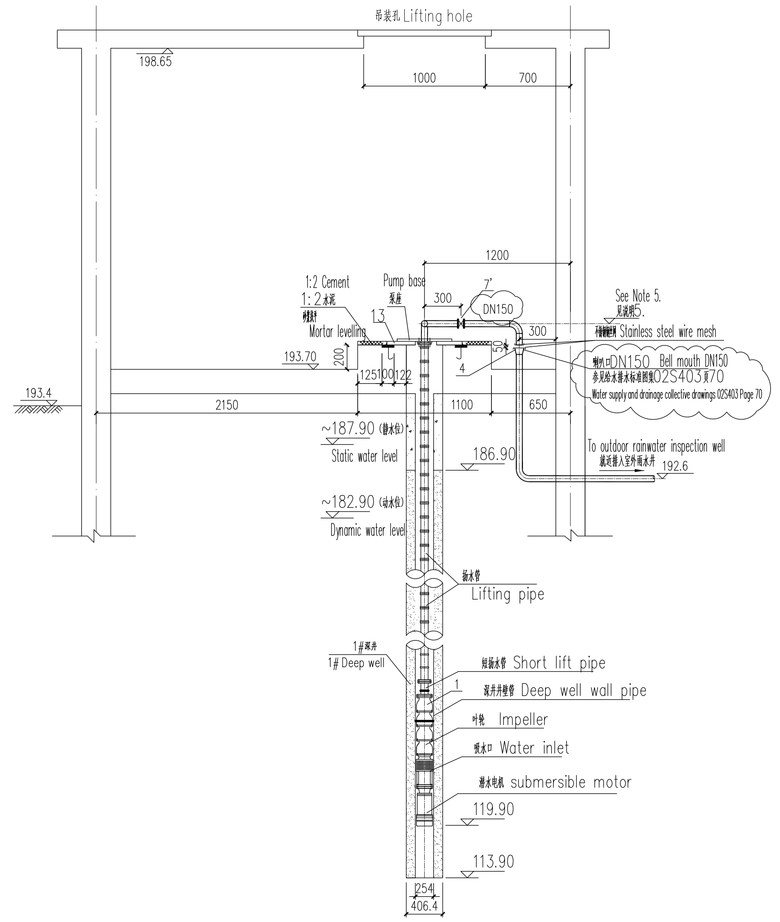
比例 Scale: 1:500 图号 DWG NO. 30-B003701S-S0204-003 版本 REV. B

A0(841x1189)



2—2 剖面图

2-2 Cross section view

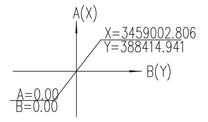


1—1 剖面图

1-1 Cross section view

说 明:

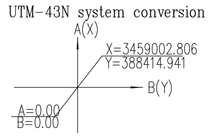
1. 图中坐标和标高单位以m计,其余均以mm计。
2. 本图采用EGM2008高程系统,图中标高为绝对标高。
   本图采用建筑(A、B)坐标,与巴基斯坦UTM-43N平面坐标系关系如下所示:



3. 深井施工单位应提供成品井,井壁管管径按ø254×10优质无缝钢管考虑,最终管径、壁厚以勘测专业为准。在实际成井后,应提供资料给设计单位核实深井潜水泵组参数。
4. 混凝土基础及环形埋铁在打井完成后先做好,钢底板待深井潜水泵到货后根据井下安装的最小空间要求确定中间开孔尺寸ø,根据泵座的要求确定螺栓位置(a)和规格(见加工简图)。钢底板加工完毕后现场与环形埋铁焊接,然后将基础四周用1:2水泥砂浆找至同钢底板顶面平。
5. 深井泵出水管中心标高,根据深井泵配带的出水口的安装高度现场确定。
6. 平时正常运行时阀门启闭状态为:阀7常开,阀7'常闭。深井潜水泵安装检修或停止运行较长时间后,再启动时须先放掉井内污染水。操作顺序为:关闭阀7,打开阀7',启动深井潜水泵,运行一段时间后见该管流出的水水质清澈后立即将各阀门恢复到正常运行状态。
7. 深井泵房内设备均需接地。
8. 由于本工程初步设计尚未审定,未通过NTDC审核,最终方案还存在修改的可能性,施工前请联系业主与设计确认。由于目前深井泵设备尚未招标,本图为根据给水排水手册设备资料暂估,深井泵设备应以最终招标资料为准,并在设备到货安装前复核基础及埋件尺寸。
9. 其他说明详见本卷册01图。

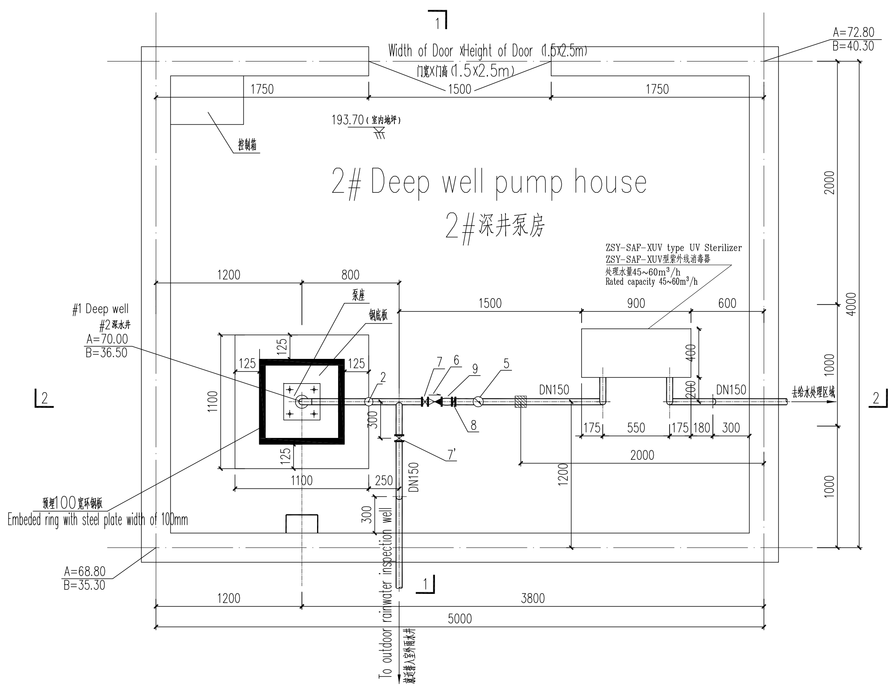
Notes:

1. Unit of coordinate and elevation is m,that of other dimensions are in mm.
2. EGM2008 elevation is adopted in this volume,elevation represents absolute elevation.The station coordinate(AB) is adopted in this volume,station coordinate and UTM-43N system conversion formula shows as follow:



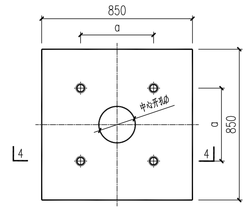
inal diameter,wall thickness.shall be based on surveying major.After deep well is completed,information of deep-well pumps shall be provided to design institute to check parameters.

4. Concrete foundation and circular embeded part shall be done first after drilling work finished,hole size of steel plate will be determined based on minimum installation space of deep-well pump,bolt location(a) and specification(b) would be determined based on pump frame requirements(fabrication skematic diagram)Steel plate fabrication,after fabricated,will be welded with cicular embeded part on site,then foundation surrounds will be made up level to to top elevation of steel plate with 1:2 cement mortar.
5. Center elevtion of outlet pipe shall be adjusted with actual elevation of autual arrived pump outlet.
6. Opening and closing position in operation : valve 7 normally open , valve 7' normally closed.If deep-well pump going through a long time of maitanance or shutdown polluted water in the well must be drawn off as pump restarting.operation order: close valve 7,then open valve 7' , start deep-well pump,recover the valves to a normal state immediately,provided water quality is clear after running for a while.
7. All equipments in deep-well pump house must be gr
8. As basic design has not been reviewed yet by NTDC,final scheme is up on air.Please contact CET and Design Institute before construction in order to make sure that scheme is right.Bid inviting of deep-well pump is not carried out,pump data refer to equipment information of water supply and drainage handbook,deep-well pump equipment shall be subject to final biding informationand review foundation and embeded size before equipment transported to the site.
9. Other descriptions refer to S0204-01 drawing of this volume.



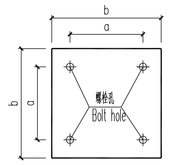
平面图

Layout Plan



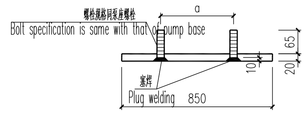
钢底板(厚20)加工简图

Processing schematic of Steel bottom plate (thickness 20)



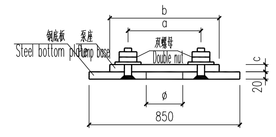
泵座示意图

Diagrammatic drawing of pump base



钢底板加工4—4剖面图

4-4 Cross section drawing for processing of steel bottom plate



泵座、钢底板连接图

Connection layout of pump base and steel bottom plate

13	Steel plate 钢底板	850x850x20
12	90° Elbow 90°弯头	DN150
11	Tee joint 三通	DN150
10	Steel Pipe lined with plastics 衬塑镀锌钢管	DN150 , PN16
9	Flexible rubber joint 可曲挠橡胶接头	DN150 , PN10
8	Butterfly valve 蝶 阀	DN150 , PN10
7(7')	Sluice valve 闸 阀	DN150 , PN10
6	Microresistance slow closing check valve 微阻缓闭止回阀	DN150 , PN10
5	Electromagnetic flowmeter 电磁流量计	DN150
4	Horn mouth 喇叭口	DN150
3	Pressure gage 压力表	Y-60,0~1.0MPa
2	Automatic air valve 自动排气阀	ARSX-0015 DN15
1	Deep-well pump 深井潜水泵组	Q=80m³/h,H=52m,380V , 18.5kW
编号 No.	名 称 Name	规 格 Parameters

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦联带至里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 4 - 0 0 4 B

	姓名 Name	日期 Date
设计 Designed	罗文瑞	2020.2.25
校核 Checked		
审核 Reviewed		
批准 Approved		

#2深井泵房布置图

#2 Deep Well Pump House Arrangement Drawing

比例 Scale: 1:500 图号 DWG NO. 30-B003701S-S0204-004 版本 REV. B

A0(841x1189)

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark
16	Horn mouth 喇叭口	DN150	Set 套	2	Q235a lining plastic Q235a衬塑	–	–	
15	Pipe support 管道支架		Set 套	2	Q235a	–	–	
14	Steel plate 钢底板	850x850x20	Set 块	2	Q235a	–	–	
13	45° Elbow 45°弯头	DN150	Set 只	10	Q235a lining plastic Q235a衬塑	–	–	
12	90° Elbow 90°弯头	DN150	Set 只	27	Q235a lining plastic Q235a衬塑	–	–	
11	Tee joint 三通	DN150/DN100	Set 只	2	Q235a lining plastic Q235a衬塑	–	–	
10	Tee joint 三通	DN150	Set 只	4	Q235a lining plastic Q235a衬塑	–	–	
9	Steel Pipe lined with plastics 衬塑镀锌钢管	DN150 , PN16	m 米	470	Q235a lining plastic Q235a衬塑	–	–	
9	Flexible rubber joint 可曲挠橡胶接头	DN150 , PN10	Set 只	2		–	–	
8	Butterfly valve 蝶 阀	DN150 , PN10	Set 个	2		–	–	
7	Sluice valve 闸 阀	DN150 , PN10	Set 个	4		–	–	
6	Microresistance slow closing check valve 微阻缓闭止回阀	DN150 , PN10	Set 个	2		–	–	
5	Electromagnetic flowmeter 电磁流量计	DN150	Set 个	2		–	–	
4	Pressure gage 压力表	Y-60,0~1.0MPa	Set 个	2		–	–	
3	Automatic air valve 自动排气阀	ARSX-0015 DN15	Set 台	2		–	–	
2	UV Sterilizer 紫外线消毒器	处理水量65~80m³/h	Set 台	2		–	–	
1	Deep-well pump 深井潜水泵组	Q=80m³/h,H=52m,380V , 18.5kW	set 台	2		–	–	工作泵包括潜水泵、电机、放水管、泵座、出口法兰、控制柜及电缆; Pump including submersible pump, motor, pipes, pump frame, outlet flange, control cabinet and cable.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 4 - 0 0 5 B

	姓名 Name	日期 Date
设计 Designed		2020.2.25
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO.

深井泵房及补充水管材料表

Bill Of Material For Deep-well Pump House And Make-up Water Pump

30-B003701S-S0204-005

版本 REV. B

A3(297x420)